Requirement Analysis

Solution Requirements (Functional & Non-Functional)

Field	Details
Team Lead	Konkimalla Bhagyasri Lasya
Team Members	Bhavitha Medakayala, Yogiswari Sanapala, Divya kauluri, Nagaram praveenya
Domain	India's Agriculture Crop Production Analysis
Tools Used	Tableau
Dataset	Historical Indian Crop Production Data, 1997–2021
Published Dashboard	https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject

Project Context

This report presents a comprehensive visual exploration of India's agricultural cultivation using 25 years (1997–2021) of historical crop production data. Nine Tableau visualizations — a KPI Summary panel, State-wise Production Map, Year-wise Total Production chart, Trend of Agricultural Yield chart, Cultivated Area by Crop chart, State-wise Cultivated Area Comparison chart, Leading States by Production chart, Production Distribution by Unit Type donut, and Season-wise Production pie — are combined into four role-specific dashboards and a five-point guided Story, enabling stakeholders to uncover patterns, identify areas of growth or concern, and make data-driven decisions.

By harnessing the power of Tableau, this report presents the data in a visually appealing, interactive form that lets readers explore the intricacies of India's agricultural cultivation for themselves.

Scenario 1: Small-Scale Farmer

Rajesh is a small-scale farmer in Uttar Pradesh who wants to maximize his crop yield and income. However, he struggles to decide which crops to grow each season due to changing weather patterns and market demand. By analyzing historical agricultural data (1997–2021), Rajesh needs insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk. The Farmer's View dashboard serves Rajesh through the Cultivated Area by Crop and Season-wise Production charts.

Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively. The Policy Analyst's View dashboard serves Ananya through the State-wise Production Map and Trend of Agricultural Yield chart.

Scenario 3: Agribusiness Investor

Vikram is an agribusiness investor looking to invest in high-growth agricultural sectors. He needs to analyze historical crop production data to identify profitable crops, emerging trends, and regions with consistent growth. By leveraging visual analytics in Tableau, Vikram seeks to minimize investment risk and make informed decisions on where and what to invest in the agricultural market. The Investor's View dashboard serves Vikram through the Leading States by Production and Cultivated Area by Crop charts.

Functional Requirements

FR #	Requirement	Description
FR-01	Data Connectivity	System shall connect to the cleaned crop production CSV dataset (345,407 rows, 1997–2021)
FR-02	State/Region Filter	System shall allow users to filter all visualizations by State Name
FR-03	Season Filter	System shall allow multi-select filtering by Season (Kharif, Rabi, Whole Year, Autumn, Summer, Winter)
FR-04	Year Range Filter	System shall provide a slider filter for Crop Year Numeric (1997–2021)
FR-05	Crop Filter	System shall allow filtering by individual crop types (57 unique crops)
FR-06	State-wise Production Map	System shall display a filled geographic map of India colored by SUM(Production) per state
FR-07	Year-wise Total Production Chart	System shall display a line chart showing total national production by year (1997–2021)
FR-08	KPI Summary Panel	System shall display key summary metrics — total production, total cultivated area, and average yield — in a KPI panel
FR-09	Season-wise Production Chart (Pie)	System shall display a pie chart comparing total production across all 6 seasons
FR-10	Cultivated Area by Crop Chart	System shall display a bar chart of total cultivated area summed by crop type
FR-11	Trend of Agricultural Yield Chart	System shall display a line chart showing average Yield (Production/Area) trended by year
FR-12	State-wise Cultivated Area Comparison Chart	System shall display a bar chart comparing total cultivated area across states
FR-13	Leading States by Production Chart	System shall display a ranked chart of the top states by total production
FR-14	Production Distribution by Unit Type Chart (Donut)	System shall display a donut chart showing the share of production by measurement unit type
FR-15	Role-Based Dashboards	System shall combine charts into dashboards: Farmer, Policy Analyst, and Investor views
FR-16	Tableau Story	System shall present a guided narrative Story with story points, one per persona

FR-17	Tableau Public Publishing	System shall be published to Tableau Public for browser-based access without login
FR-18	Tooltip on Hover	All charts shall display State Name and Production value on hover

Non-Functional Requirements

NFR #	Category	Requirement
NFR-01	Usability	Dashboard UI shall be navigable by non-technical users (farmers, policy makers) without training
NFR-02	Performance	All filters shall respond and re-render within 3 seconds on the full 345K row dataset
NFR-03	Reliability	Published Tableau Public workbook shall be available 99%+ of the time via public URL
NFR-04	Scalability	Solution architecture shall support addition of new years' data without redesigning dashboards
NFR-05	Availability	Dashboard shall be accessible via standard web browsers (Chrome, Edge, Safari) without plugins
NFR-06	Security	No personally identifiable information (PII) is stored or displayed; dataset is fully public
NFR-07	Maintainability	Workbook shall be saved as .twbx (packaged workbook) with embedded data for portability
NFR-08	Compatibility	Solution shall render correctly on desktop (1366×768 or higher) and Tableau Public mobile view